

Residual Trend and Mean-Reversion Extension to the Avellaneda–Lee Daily ETF Replication

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1 Motivation

The baseline replication plan in `replication_plan.tex` follows the Avellaneda–Lee ETF residual strategy: stock returns are decomposed into a sector ETF component and an idiosyncratic residual, and trades are opened when the residual level is sufficiently far from its estimated equilibrium. This is a pure contrarian interpretation of residual displacement. If a stock appears expensive relative to its sector ETF, the baseline sells the stock and buys the ETF hedge; if it appears cheap, the baseline buys the stock and sells the ETF hedge.

That assumption is natural for a paper replication, but it is restrictive in post-2020 data. The COVID shock, the rate-hiking cycle, and the artificial intelligence repricing created persistent relative trends inside sectors. In such environments, a residual that keeps rising may not be temporary noise; it may reflect a sustained idiosyncratic repricing. A strategy that mechanically fades every large residual can repeatedly short relative winners too early.

This extension keeps the paper’s residual framework but changes the trading decision. The residual is allowed to be either a mean-reverting displacement or a relative trend. The strategy first asks which structure is more plausible, and only then decides whether to fade or ride the residual.

2 Residual Framework

For stock i assigned to sector ETF $I(i)$, daily returns are decomposed as

$$R_{i,t} = \beta_{i,t} R_{I(i),t} + \tilde{R}_{i,t}.$$

The coefficient $\beta_{i,t}$ is estimated from a trailing window using only information available before the trading decision. As in the baseline plan, the cumulative residual level is

$$X_{i,t} = \sum_{u \leq t} \tilde{R}_{i,u}.$$

Large positive $X_{i,t}$ means the stock has cumulatively outperformed its sector ETF after beta adjustment. Large negative $X_{i,t}$ means it has underperformed.

The baseline strategy assumes that $X_{i,t}$ is stationary and mean-reverting. This extension treats stationarity as one possible state rather than a permanent assumption.

3 Two Residual States

3.1 Mean-Reversion State

In the mean-reversion state, the residual level is interpreted as a temporary relative-value displacement. The OU score from the baseline plan remains the decision variable:

$$s_{i,t}^{MR} = \frac{X_{i,t} - \hat{m}_{i,t}}{\hat{\sigma}_{eq,i,t}}.$$

A high positive score indicates that the stock is expensive relative to its sector ETF, so the mean-reversion trade is short stock and long ETF. A high negative score indicates that the stock is cheap relative to its sector ETF, so the trade is long stock and short ETF.

3.2 Trend State

In the trend state, a persistent residual move is treated as information rather than as a mispricing to fade. Define a trailing residual trend score over lookback L :

$$s_{i,t}^{TR} = \frac{X_{i,t} - X_{i,t-L}}{\hat{\sigma}_{\Delta X,i,t} \sqrt{L}},$$

where $\hat{\sigma}_{\Delta X,i,t}$ is a trailing volatility estimate of daily residual increments. A positive trend score means the stock is persistently outperforming its sector ETF. A negative trend score means it is persistently underperforming.

The trend trade follows the residual:

$$\begin{aligned} s_{i,t}^{TR} > c_{TR} &\Rightarrow \text{long stock, short ETF,} \\ s_{i,t}^{TR} < -c_{TR} &\Rightarrow \text{short stock, long ETF.} \end{aligned}$$

This is still a relative-value trade. It does not abandon sector hedging; it rides idiosyncratic momentum against the sector benchmark.

4 Regime Decision

The central extension is a mode selection rule. At each decision date, the strategy chooses between:

$$q_{i,t}^{MR} = -\text{sign}(s_{i,t}^{MR}) \quad \text{when } |s_{i,t}^{MR}| \text{ is large and trend is weak,}$$

and

$$q_{i,t}^{TR} = \text{sign}(s_{i,t}^{TR}) \quad \text{when } |s_{i,t}^{TR}| \text{ is large and trend is confirmed.}$$

Here $q_{i,t}$ is the stock-side direction: +1 means long stock and -1 means short stock. A simple hard-switching rule is

$$q_{i,t} = \begin{cases} \text{sign}(s_{i,t}^{TR}), & |s_{i,t}^{TR}| \geq c_{TR}, \\ -\text{sign}(s_{i,t}^{MR}), & |s_{i,t}^{TR}| < c_{TR} \text{ and } |s_{i,t}^{MR}| \geq c_{MR}, \\ 0, & \text{otherwise.} \end{cases}$$

The trend condition is evaluated first. This priority reflects the empirical problem that motivates the extension: in strong post-2020 repricing regimes, the costly error is often fading a persistent residual too early.

5 Volume and Information Confirmation

The source paper reports that ETF residual strategies improve when volume is used to distinguish low-volume noise from high-volume information. This idea fits naturally into the trend-versus-mean-reversion extension.

Let

$$\nu_{i,t} = \frac{V_{i,t}}{\overline{V}_{i,t}^{(60)}}$$

be relative volume. A residual move on unusually high volume is treated as more likely to contain information. A residual move on ordinary or low volume is more likely to be a temporary dislocation.

One implementation is to require trend confirmation:

$$|s_{i,t}^{TR}| \geq c_{TR} \quad \text{and} \quad \nu_{i,t} \geq c_V.$$

Another implementation is to form a trading-time residual increment:

$$\tilde{R}_{i,t}^{TT} = \frac{\tilde{R}_{i,t}}{\nu_{i,t}^\phi}, \quad \phi \geq 0.$$

This reduces the influence of high-volume residual jumps in the mean-reversion fit, making the strategy less likely to fade moves that appear to be information-driven.

6 Portfolio Construction

The portfolio construction remains close to the baseline replication. For stock-side signal $q_{i,t} \in \{-1, 0, +1\}$ and base stock allocation Λ_t , the stock weight is

$$w_{i,t} = \Lambda_t q_{i,t}.$$

The sector ETF hedge is

$$w_{I(j),t} = - \sum_{i:I(i)=I(j)} \hat{\beta}_{i,t} w_{i,t}.$$

Thus both residual trend trades and residual mean-reversion trades remain sector-neutral in the same sense as the paper's ETF strategy. The difference is the sign of the stock-side trade, not the definition of factor neutrality.

Net daily PNL is still

$$\pi_t = w_t^\top R_t - c \|w_t - w_{t-1}\|_1,$$

with transaction costs charged on position changes.

7 Diagnostics

The extension must report performance by trading mode. Aggregate PNL is not enough, because the key research question is whether residual trend and residual mean reversion work in different states.

For each date and stock, trades should be labeled as one of:

$$\{\text{trend long, trend short, mean-reversion long, mean-reversion short, flat}\}.$$

The backtest should then report:

- cumulative PNL by mode;
- hit rate by mode;
- average holding period by mode;
- turnover and transaction costs by mode;
- sector contribution by mode; and
- exposure to the assigned ETF after hedging.

If trend-mode PNL dominates, the post-2020 market may be better described by relative momentum than by residual convergence. If mean-reversion-mode PNL dominates only in calm regimes, then the strategy should gate contrarian trades more aggressively. If neither mode works after costs, the residual signal itself is not strong enough for the chosen universe and horizon.

8 Implementation Sequence

This note should be implemented after the baseline assigned-sector-ETF replication is working. The proposed sequence is:

1. keep the assigned sector ETF residual construction from `replication.plan.tex`;
2. compute residual levels $X_{i,t}$ and OU s-scores as before;
3. add residual trend scores over one or more fixed lookbacks;
4. add mode labels for trend, mean reversion, and flat states;
5. trade with trend priority, then compare against pure mean reversion;
6. add volume confirmation or trading-time residuals; and
7. report PNL separately by mode before optimizing thresholds.

The purpose is not to data-mine a better backtest. The purpose is to test a specific post-2020 hypothesis: some large residuals represent temporary relative-value dislocations, while others represent persistent idiosyncratic repricing that should be followed rather than faded.